Task on VPC

1. **Create one VPC, with 1 public subnet and 1 private subnet.**

Step 1

1. Login to your AWS Management Console.
2. Search for VPC in the search bar.
3. Open the VPC Dashboard.
4. Click Your VPCs.
5. Click Create VPC.
6. Select VPC only.
7. Enter the following details:

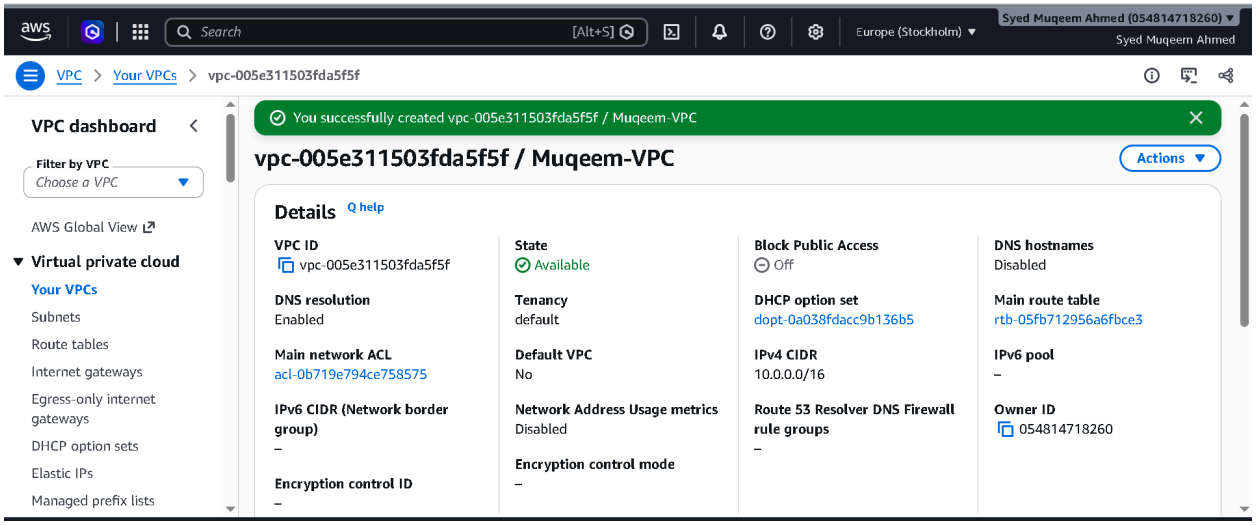
Name tag: Muqeem-VPC

IPv4 CIDR: 10.0.0.0/16

IPv6 CIDR: No IPv6 CIDR block

Tenancy: Default

8. Click Create VPC.



Step 2: Create the Public Subnet

1. Open Subnets from the left navigation panel.

2. Click Create subnet.

3. Select your VPC: Muqeem-VPC.

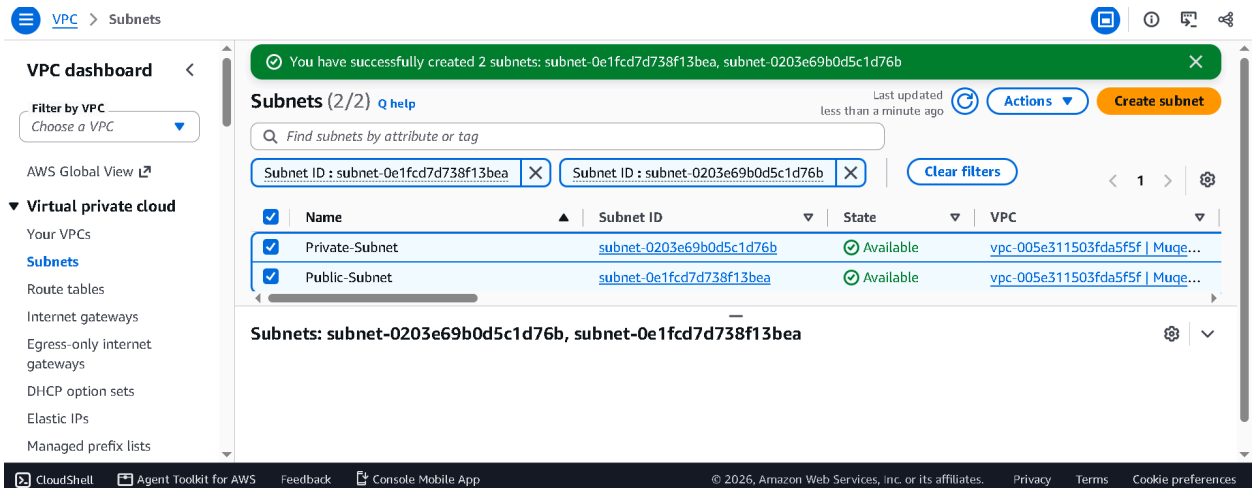
Subnet name: Public-Subnet

Availability Zone: Select any one

IPv4 subnet CIDR: 10.0.1.0/24

follow the same process for private subnet

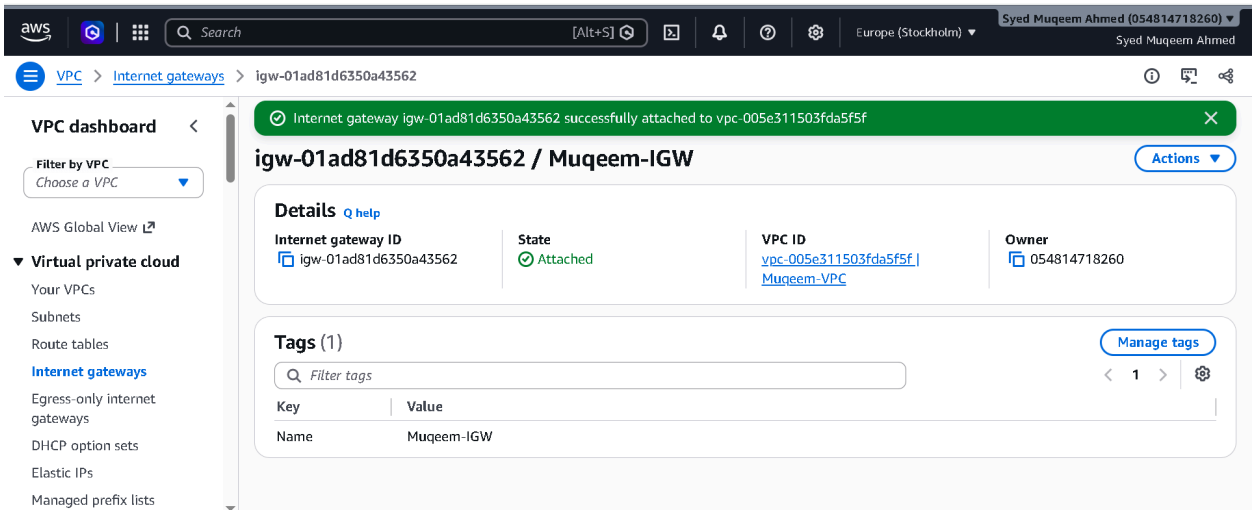
5. Click Create subnet



Step 4: Create an Internet Gateway (IGW)

An Internet Gateway allows resources in your public subnet to communicate with the internet.

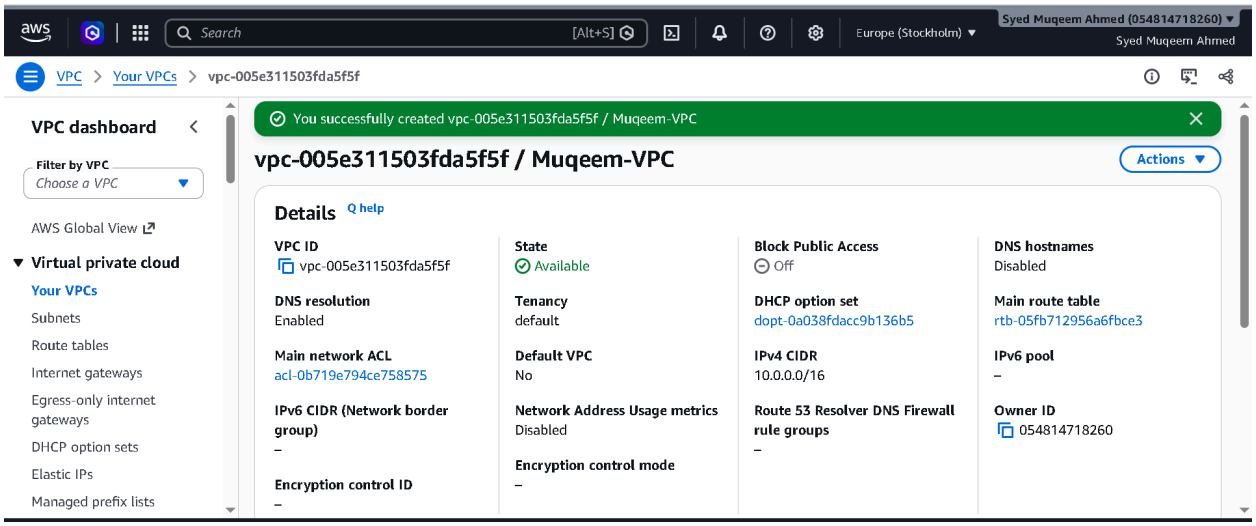
1. Open Internet Gateways from the left panel.
2. Click Create internet gateway.
3. Enter the name: Muqeem-IGW.
4. Click Create internet gateway.
5. Select the newly created Internet Gateway.
6. Click Actions → Attach to a VPC.
7. Select Muqeem-VPC.
8. Click Attach internet gateway.



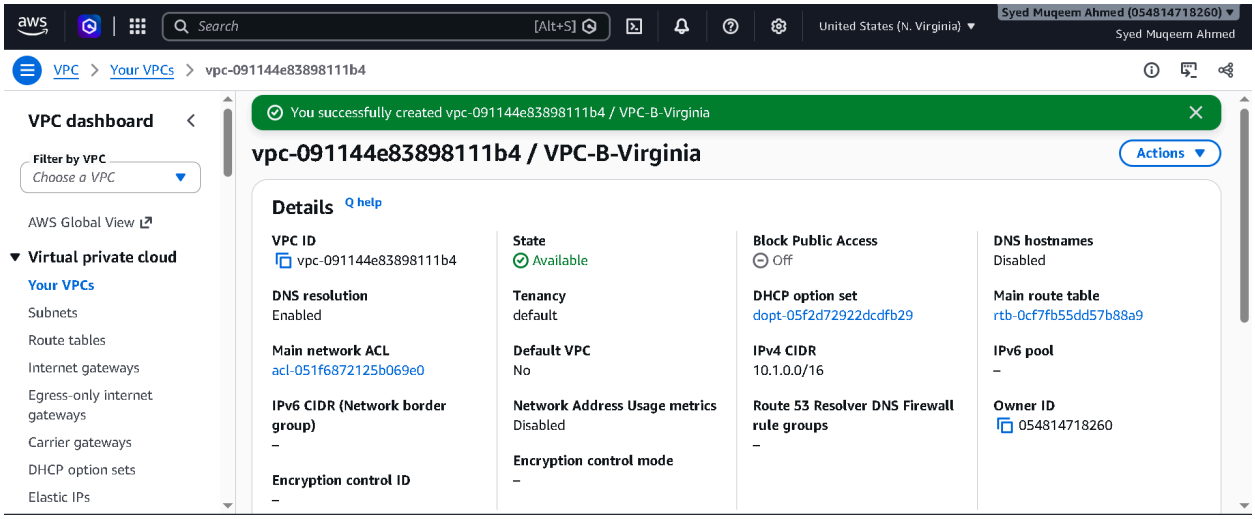
Step 5: Create a Public Route Table

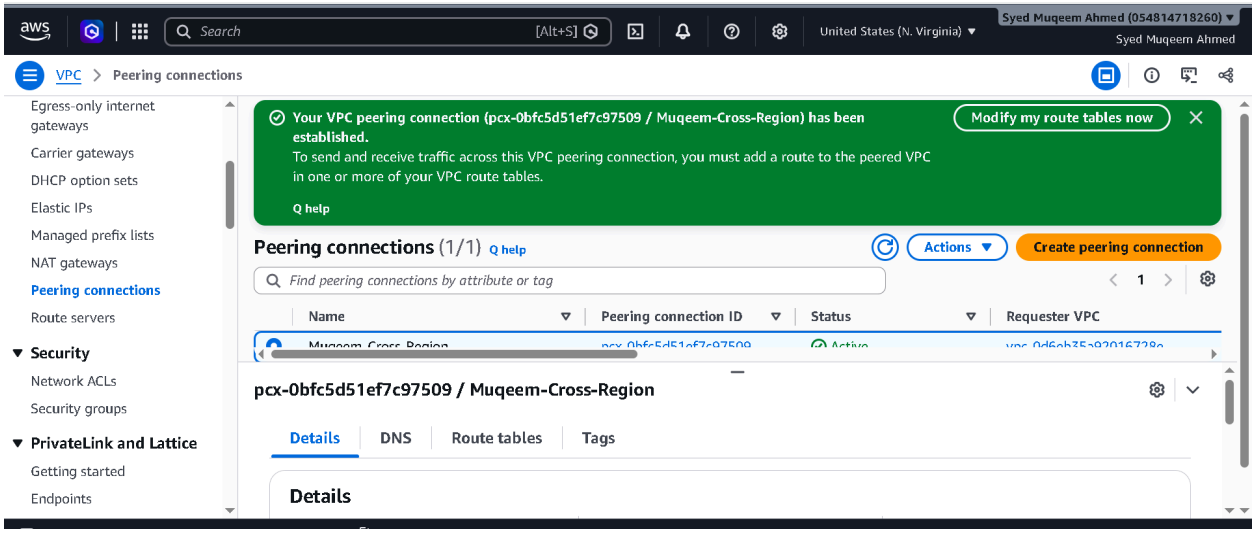
2. Enable VPC peering for cross-region.

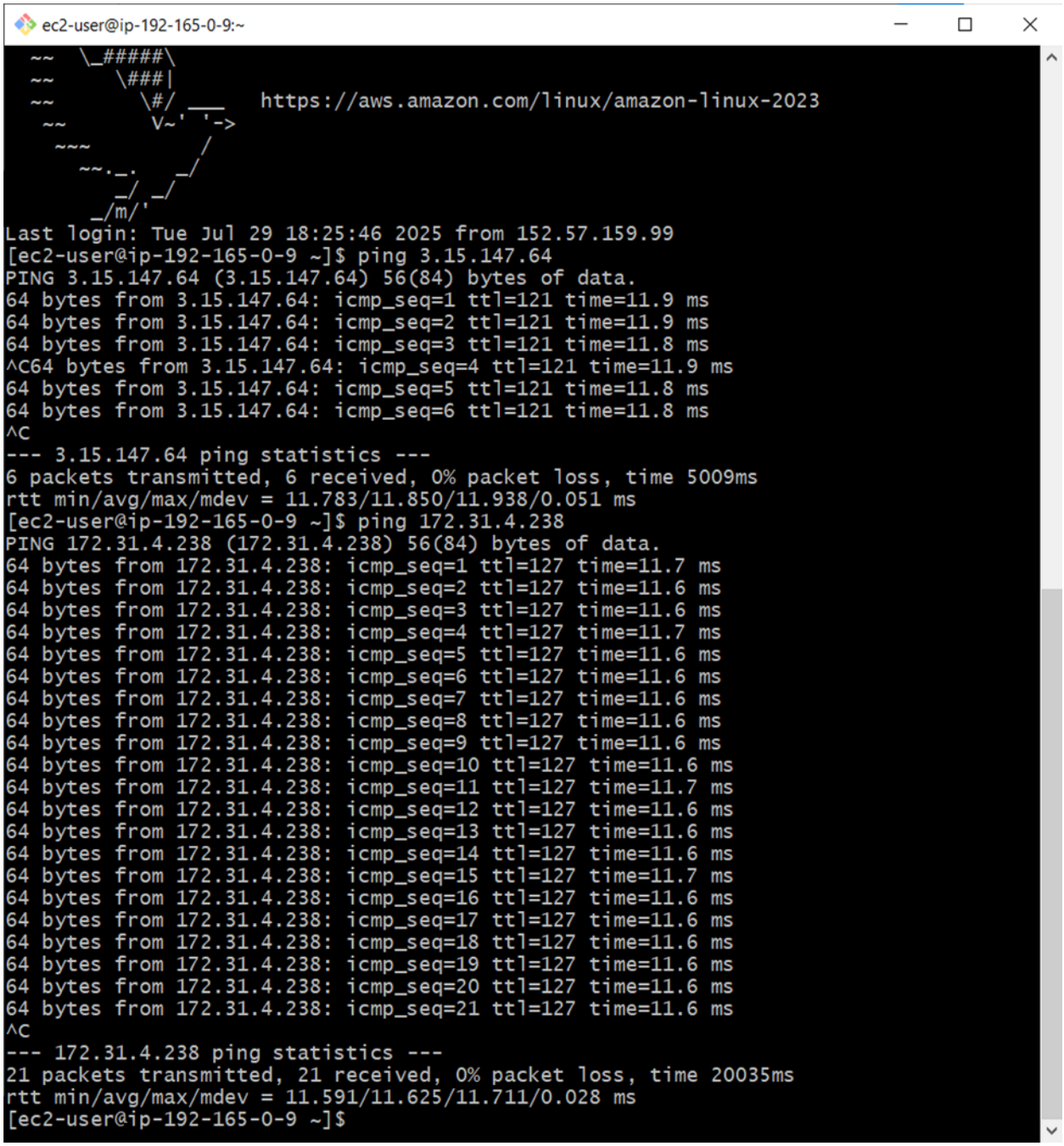
#VPC-1



#VPC-2


Search
[Alt+S]
United States (N. Virginia)
Syed Muqeem Ahmed (054814718260)
Syed Muqeem Ahmed
VPC > Your VPCs > vpc-091144e83898111b4
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
You successfully created vpc-091144e83898111b4 / VPC-B-Virginia
vpc-091144e83898111b4 / VPC-B-Virginia
Actions
Details
Q help
VPC ID
vpc-091144e83898111b4
State
Available
Block Public Access
Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-05f2d72922dcdfb29
Main route table
rtb-0cf7fb55dd57b88a9
Main network ACL
acl-051f6872125b069e0
Default VPC
No
IPv4 CIDR
10.1.0.0/16
IPv6 pool
–
IPv6 CIDR (Network border group)
–
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
–
Owner ID
054814718260
Encryption control ID
–
Encryption control mode
–


Search
[Alt+S]
United States (N. Virginia)
Syed Muqeem Ahmed (054814718260)
Syed Muqeem Ahmed
VPC > Peering connections
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Your VPC peering connection (pcx-0bfc5d51ef7c97509 / Muqeem-Cross-Region) has been established.
To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.
Q help
Modify my route tables now
Peering connections (1/1)
Q help
Actions
Create peering connection
Find peering connections by attribute or tag
Name
Peering connection ID
Status
Requester VPC
pcx-0bfc5d51ef7c97509 / Muqeem-Cross-Region
Details
DNS
Route tables
Tags
Details



3. Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

Steps:

1.Created one instance with selecting my-vpc.

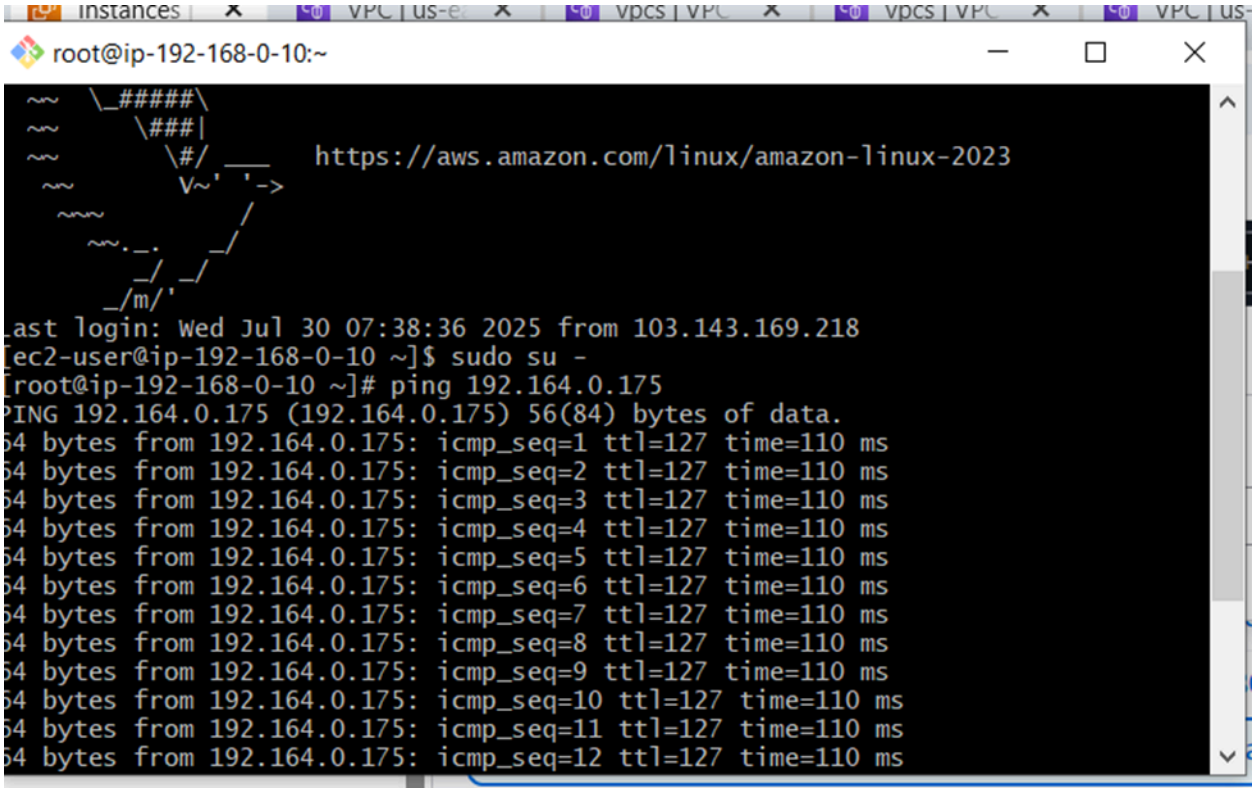
2.Connected to my pub-subnet

3.Created a peer connection with requester my-vpc and accepter yunus aws account id and his vpc id and his region

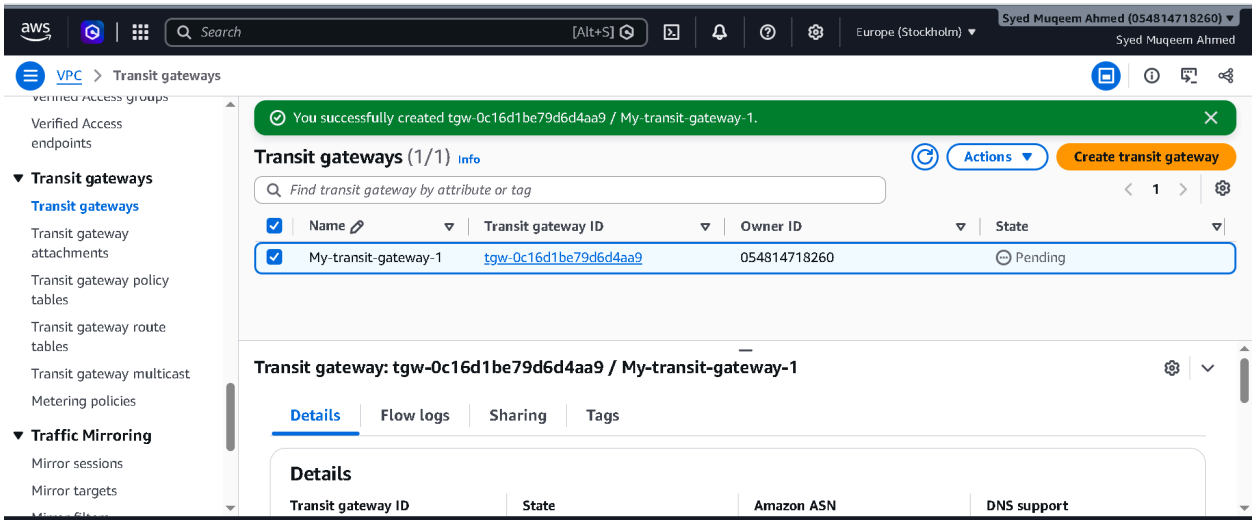
4.In route tables entered the yunus cidr range 192.164.0.0/20 in target selected peering connection.

5.Once the accepter accepted the request he also edited his route with my CIDR range and selected peering connection.

6.Now able to ping each other.



4. Set up a VPC Transit Gateway.



5. Set up a VPC Endpoint.

Step 1: Open VPC

1. Log in to your AWS Console.
2. Search for VPC.
3. Open the VPC Dashboard.
4. From the left-side menu, select Endpoints.
5. Click Create endpoint.

Step 2: Configure the endpoint

Name tag: Muqeem-endpoint-s3

Service category: AWS services

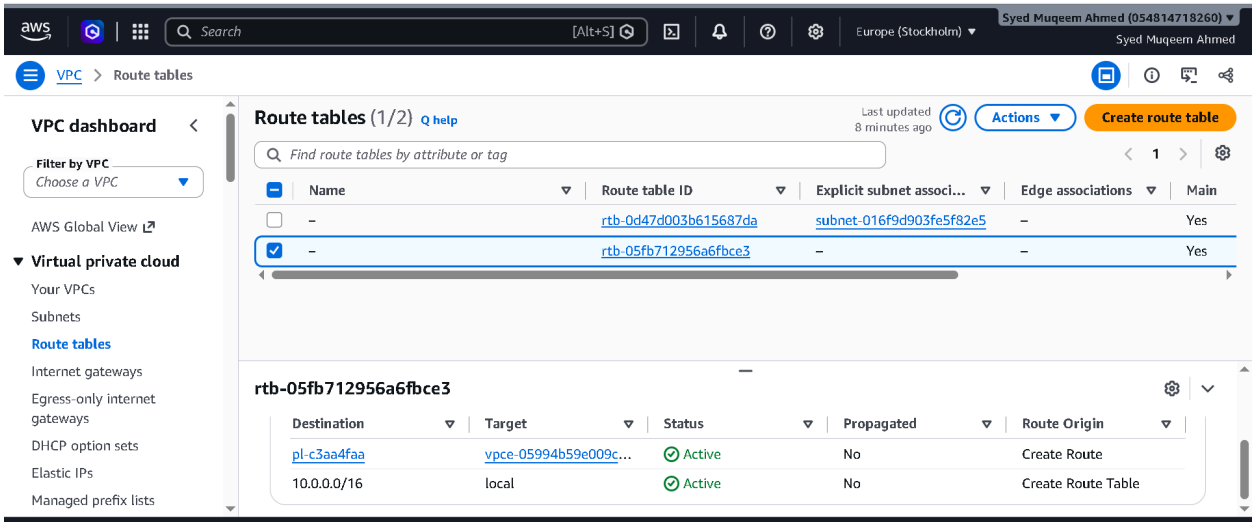
Service name: Search for “S3” and select

Type: Gateway

VPC: Select your existing VPC

Step 3: Select Route Tables

1. Scroll down to the Route tables section.
2. Select your Private Route Table.
3. Make sure you select the route table associated with your private subnet.



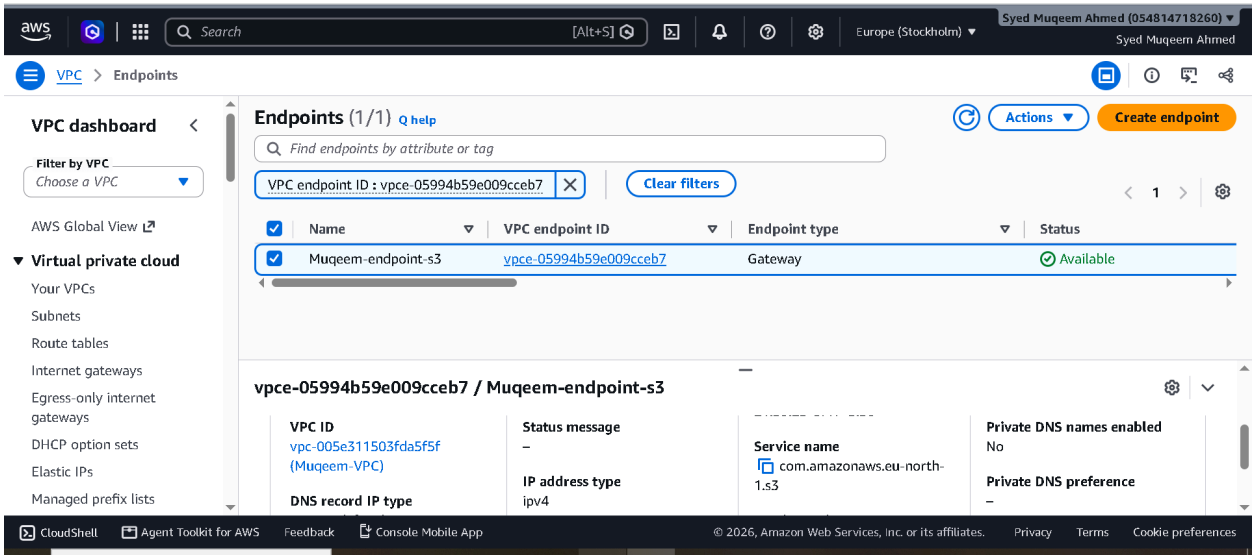
4. Verify the VPC Endpoint

After creating the endpoint:

1. Go to VPC → Endpoints.

2. Select S3-VPC-Endpoint.

3. Check its status is Available.

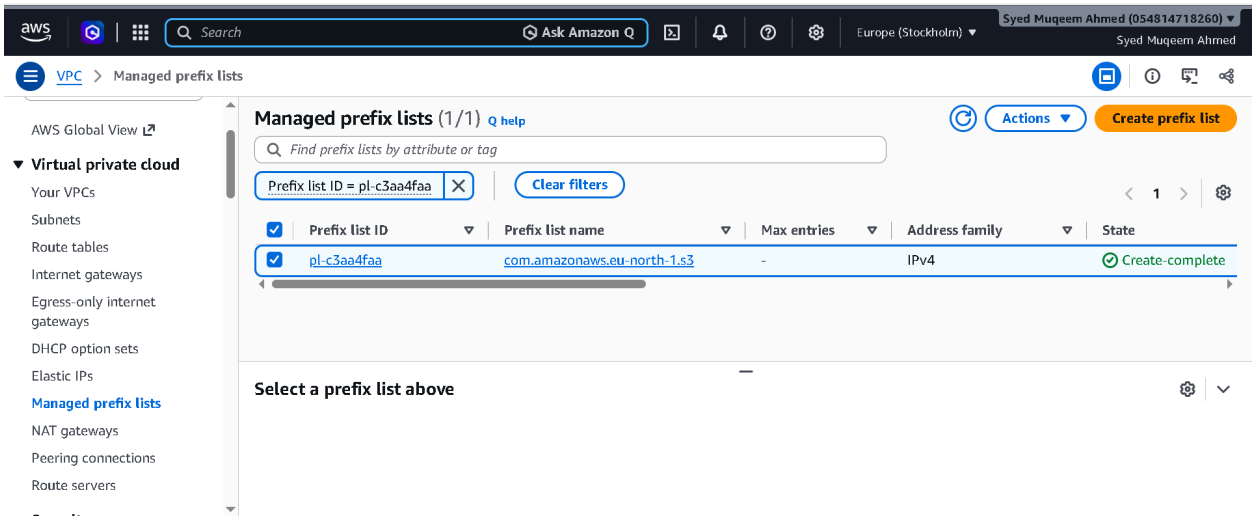


4. Open your private route table.

5. Go to the Routes tab.

You should see a new route similar to:
Destination Target
pl-xxxxxxxx (S3 prefix list) vpce-xxxxxxxx

This confirms that your private route table has a route to S3 through the VPC Endpoint.


Search
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Europe (Stockholm)
Syed Muqeem Ahmed (054814718260)
Syed Muqeem Ahmed
VPC
Managed prefix lists
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Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Managed prefix lists (1/1)
Q help
Actions
Create prefix list
Find prefix lists by attribute or tag
Prefix list ID = pl-c3aa4faa
Clear filters
1
Prefix list ID
Prefix list name
Max entries
Address family
State
pl-c3aa4faa
com.amazonaws.eu-north-1.s3
-
IPv4
Create-complete
Select a prefix list above